Assignment 1

1. create a generic Range<T> class that represents a range of values from a minimum value to a maximum value. The range should support basic operations such as checking if a value is within the range and determining the length of the range.
   Requirements:
   1. Create a generic class named Range<T> where T represents the type of values.
   2. Implement a constructor that takes the minimum and maximum values to define the range.
   3. Implement a method IsInRange(T value) that returns true if the given value is within the range, otherwise false.
   4. Implement a method Length() that returns the length of the range (the difference between the maximum and minimum values).
   5. Note: You can assume that the type T used in the Range<T> class implements the IComparable<T> interface to allow for comparisons.

2. You are given a list of integers. Your task is to find and return a new list containing only the even numbers from the given list.

3. implement a custom list called **FixedSizeList<T>** with a predetermined capacity. This list should not allow more elements than its capacity and should provide clear messages if one tries to exceed it or access invalid indices.
   Requirements:
   1. Create a generic class named **FixedSizeList<T>**.
   2. Implement a constructor that takes the fixed capacity of the list as a parameter.
   3. Implement an **Add** method that adds an element to the list, but throws an exception if the list is already full.
   4. Implement a **Get** method that retrieves an element at a specific index in the list but throws an exception for invalid indices.

4. Given a string, find the first non-repeated character in it and return its index. If there is no such character, return -1. Hint you can use dictionary